

01

Relations

1. CARTESIAN PRODUCT OF TWO SETS:

Given two sets A & B . The cartesian product $A \times B$ is the set of all ordered pairs of the form (a, b) where the first entry comes from set A & second comes from set B .

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

e.g. $A = \{1, 2, 3\}$

$$B = \{p, q\}$$

$$A \times B = \{(1, p), (1, q), (2, p), (2, q), (3, p), (3, q)\}$$

Note: (i) If either A or B is the null set, then $A \times B$ will also be empty set, i.e. $A \times B = \phi$

(ii) If $n(A) = p, n(B) = q$, then $n(A \times B) = pq$, where $n(X)$ denotes the number of elements in set X .

Illustration 1

If $A \times B = \{(p, x); (p, y); (q, x); (q, y)\}$, find A and B .

Solution:

A is a set of all first entries in ordered pairs in $A \times B$. B is a set of all second entries in ordered pairs in $A \times B$. Thus $A = \{p, q\}$ and $B = \{x, y\}$

Illustration 2

If A and B are two sets, and $A \times B$ consists of 6 elements: If three elements of $A \times B$ are $(2, 5)$ $(3, 7)$ $(4, 7)$ find $A \times B$.

Solution:

Since, $(2, 5)$ $(3, 7)$ and $(4, 7)$ are elements of $A \times B$. So, we can say that 2, 3, 4 are the elements of A and 5, 7 are the elements of B

So, $A = \{2, 3, 4\}$ and $B = \{5, 7\}$

Now, $A \times B = \{(2, 5); (2, 7); (3, 5); (3, 7); (4, 5); (4, 7)\}$

Thus, $A \times B$ contain six ordered pairs.

2. RELATION:

Let A and B be two sets. Then a relation R from A to B is a subset of $A \times B$.

Thus, R is a relation from A to $B \Leftrightarrow R \subseteq A \times B$.

Note: A Relation R from set A to B is any subset of $A \times B$. If ARB & $(a, b) \in R$ then b is image of a under R and a is preimage of b under R .

Ex. If $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$, then $R = \{(1, b), (2, c), (1, a), (3, a)\}$ being a subset of $A \times B$, is a relation from A to B . Here $(1, b), (2, c), (1, a)$ and $(3, a) \in R$, so we write $1Rb, 2Rc, 1Ra$ and $3Ra$. But $(2, b) \notin R$, so we write $2 \not R b$.

(a) Total Number of Relations:

Let A and B be two finite sets consisting of m and n elements respectively. Then $A \times B$ consists of mn ordered pairs. **So, total number of subsets of $A \times B$ is 2^{mn} .**

(b) Domain and Range of a relation:

Let R be a relation from a set A to a set B . Then the set of all first components or coordinates of the ordered pairs belonging to R is called to domain of R , while the set of all second components or coordinates of the ordered pairs in R is called the range of R .

Thus, $\text{Domain}(R) = \{a : (a, b) \in R\}$

and, $\text{Range}(R) = \{b : (a, b) \in R\}$

It is evident from the definition that the domain of a relation from A to B is a subset of A and its range is a subset of B .

Ex. Let $A = \{1, 3, 5, 7\}$ and $B = \{2, 4, 6, 8\}$ be two sets and let R be a relation from A to B defined by the phrase " $(x, y) \in R \Leftrightarrow x > y$ ". Under this relation R , we have $3R2, 5R2, 5R4, 7R2, 7R4$ and $7R6$
i.e. $R = \{(3, 2), (5, 2), (5, 4), (7, 2), (7, 4), (7, 6)\}$
 $\therefore \text{Dom}(R) = \{3, 5, 7\}$ and $\text{Range}(R) = \{2, 4, 6\}$

(c) Inverse Relation:

Let A, B be two sets and let R be a relation from a set A to a set B . Then the inverse of R , denoted by R^{-1} , is a relation from B to A and is defined by

$$R^{-1} = \{(b, a) : (a, b) \in R\}$$

Clearly, $(a, b) \in R \Leftrightarrow (b, a) \in R^{-1}$

Also, $\text{Dom}(R) = \text{Range}(R^{-1})$ and $\text{Range}(R) = \text{Dom}(R^{-1})$

Illustration 3

Let A be the set of first ten natural numbers and let R be a relation on A defined by $(x, y) \in R \Leftrightarrow x + 2y = 10$, i.e. $R = \{(x, y) : x \in A, y \in A \text{ and } x + 2y = 10\}$. Express R and R^{-1} as sets of ordered pairs. Determine also (i) domain of R and R^{-1} (ii) range of R and R^{-1}

Solution:

We have $(x, y) \in R \Leftrightarrow x + 2y = 10 \Leftrightarrow y = \frac{10-x}{2}, x, y \in A$

where $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

Now, $x = 1 \Rightarrow y = \frac{10-1}{2} = \frac{9}{2} \notin A$.

This shows that 1 is not related to any element in A . Similarly, we can observe. that 3, 5, 7, 9 and 10 are not related to any element of A under the defined relation

Further we find that:

For $x = 2, y = \frac{10-2}{2} = 4 \in A \quad \therefore (2, 4) \in R$

For $x = 4, y = \frac{10-4}{2} = 3 \in A \quad \therefore (4, 3) \in R$

For $x = 6, y = \frac{10-6}{2} = 2 \in A \quad \therefore (6, 2) \in R$

For $x = 8, y = \frac{10-8}{2} = 1 \in A \quad \therefore (8, 1) \in R$

Thus, $R = \{(2, 4), (4, 3), (6, 2), (8, 1)\}$

$\Rightarrow R^{-1} = \{(4, 2), (3, 4), (2, 6), (1, 8)\}$

Clearly, $\text{Dom}(R) = \{2, 4, 6, 8\} = \text{Range}(R^{-1})$

and, $\text{Range}(R) = \{4, 3, 2, 1\} = \text{Dom}(R^{-1})$

3. TYPES OF RELATION:

In this section we intend to define various types of relations on a given set A .

(a) Void Relation:

Let A be a set. Then $\phi \subseteq A \times A$ and so it is a relation on A . This relation is called the void or empty relation on A .

(b) Universal Relation:

Let A be a set. Then $A \times A \subseteq A \times A$ and so it is a relation on A . This relation is called the universal relation on A .

(c) Identity Relation:

Let A be a set. Then the relation $I_A = \{(a, a) : a \in A\}$ on A is called the identity relation on A . In other words, a relation I_A on A is called the identity relation if every element of A is related to itself only.

Ex. The relation $I_A = \{(1, 1), (2, 2), (3, 3)\}$ is the identity relation on set $A = \{1, 2, 3\}$. But relations $R_1 = \{(1, 1), (2, 2)\}$ and $R_2 = \{(1, 1), (2, 2), (3, 3), (1, 3)\}$ are not identity relations on A , because $(3, 3) \notin R_1$ and in R_2 element 1 is related to elements 1 and 3.

(d) Reflexive Relation:

A relation R on a set A is said to be reflexive if every element of A is related to itself. Thus, R on a set A is not reflexive if there exists an element $A \in A$ such that $(a, a) \notin R$.

Ex. Let $A = \{1, 2, 3\}$ be a set. Then $R = \{(1, 1), (2, 2), (3, 3), (1, 3), (2, 1)\}$ is a reflexive relation on A . But $R_1 = \{(1, 1), (3, 3), (2, 1), (3, 2)\}$ is not a reflexive relation on A , because $2 \in A$ but $(2, 2) \notin R_1$.

Note : Every Identity relation is reflexive but every reflexive relation is not identity.

(e) Symmetric Relation:

A relation R on a set A is said to be a symmetric relation iff

$$(a, b) \in R \Rightarrow (b, a) \in R \text{ for all } a, b \in A$$

i.e. $a R b \Rightarrow b R a$ for all $a, b \in A$.

Ex. Let L be the set of all lines in a plane and let R be a relation defined on L by the rule $(x, y) \in R \Leftrightarrow x$ is perpendicular to y . Then R is a symmetric relation on L , because $L_1 \perp L_2 \Rightarrow L_2 \perp L_1$
i.e. $(L_1, L_2) \in R \Rightarrow (L_2, L_1) \in R$.

Ex. Let $A = \{1, 2, 3, 4\}$ and let R_1 and R_2 be relation on A given by $R_1 = \{(1, 3), (1, 4), (3, 1), (2, 2), (4, 1)\}$ and $R_2 = \{(1, 1), (2, 2), (3, 3), (1, 3)\}$. Clearly, R_1 is a symmetric relation on A . However, R_2 is not so, because $(1, 3) \in R_2$ but $(3, 1) \notin R_2$.

(f) Transitive Relation:

Let A be any set. A relation R on A is said to be a transitive relation iff

$$(a, b) \in R \text{ and } (b, c) \in R \Rightarrow (a, c) \in R \text{ for all } a, b, c \in A$$

i.e. $a R b$ and $b R c \Rightarrow a R c$ for all $a, b, c \in A$

Ex. On the set N of natural numbers, the relation R defined by $x R y \Rightarrow x$ is less than y is transitive, because for any $x, y, z \in N$

$$x < y \text{ and } y < z \Rightarrow x < z \Rightarrow x R y \text{ and } y R z \Rightarrow x R z$$

Ex. Let L be the set of all straight lines in a plane. Then the relation 'is parallel to' on L is a transitive relation, because from any $\ell_1, \ell_2, \ell_3 \in L$.

$$\ell_1 \parallel \ell_2 \text{ and } \ell_2 \parallel \ell_3 \Rightarrow \ell_1 \parallel \ell_3$$

(g) Asymmetric Relation Definition:

In Set theory, A relation R on a set A is known as asymmetric relation if no $(b, a) \in R$ when $(a, b) \in R$ or we can even say that relation R on set A is asymmetric if only if $(a, b) \in R \Rightarrow (b, a) \notin R$.

For example: If R is a relation on set $A = \{18, 9\}$ then $(9, 18) \in R$ indicates $18 > 9$ but $(18, 9) \notin R$, Since 9 is not greater than 18.

Note- Asymmetric relation is the opposite of symmetric relation but not considered as equivalent to antisymmetric relation.

(h) Antisymmetric Relation:

Let A be any set. A relation R on set A is said to be an antisymmetric relation iff $(a, b) \in R$ and $(b, a) \in R \Rightarrow a = b$ for all $a, b \in A$

Ex. Let R be a relation on the set N of natural numbers defined by $x R y \Leftrightarrow 'x \text{ divides } y'$ for all $x, y \in N$

This relation is an antisymmetric relation on N . Since for any two numbers $a, b \in N$

$a|b$ and $b|a \Rightarrow a = b$ i.e. $a R b$ and $b R a \Rightarrow a = b$

(i) Equivalence Relation:

A relation R on a set A is said to be an equivalence relation on A iff

(i) It is reflexive i.e. $(a, a) \in R$ for all $a \in A$

(ii) It is symmetric i.e. $(a, b) \in R \Rightarrow (b, a) \in R$ for all $a, b \in A$

(iii) It is transitive i.e. $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$ for all $a, b, c \in A$.

Note : It is not necessary that every relation which is symmetric and transitive is also reflexive.

Illustration 4

Three relation R_1, R_2 and R_3 are defined on set $A = \{a, b, c\}$ as follows:

(i) $R_1 \{ (a, a), (a, b), (a, c), (b, b), (b, c), (c, a), (c, b), (c, c) \}$

(ii) $R_2 \{ (a, b), (b, a), (a, c), (c, a) \}$

(iii) $R_3 \{ (a, b), (b, c), (c, a) \}$

Find whether each of R_1, R_2 and R_3 is reflexive, symmetric and transitive.

Solution:

(i) Reflexive: Clearly, $(a, a), (b, b), (c, c) \in R_1$. So, R_1 is reflexive on A .

Symmetric: We observe that $(a, b) \in R_1$ but $(b, a) \notin R_1$. So, R_1 is not symmetric on A .

Transitive: We find that $(b, c) \in R_1$ and $(c, a) \in R_1$ but $(b, a) \notin R_1$. So, R is not transitive on A .

(ii) Reflexive: Since $(a, a), (b, b)$ and (c, c) are not in R_2 . So, it is not a reflexive relation on A .

Symmetric: We find that the ordered pairs obtained by interchanging the components of ordered pairs in R_2 are also in R_2 . So, R_2 is a symmetric relation on A .

Transitive: Clearly $(c, a) \in R_2$ and $(a, b) \in R_2$ but $(c, b) \notin R_2$. So, it is not a transitive relation on R_2 .

(iii) Reflexive: Since none of $(a, a), (b, b)$ and (c, c) is an element of R_3 . So, R_3 is not reflexive on A .

Symmetric: Clearly, $(b, c) \in R_3$ but $(c, b) \notin R_3$. so, is not symmetric on A .

Transitive: Clearly, $(b, c) \in R_3$ and $(c, a) \in R_3$ but $(b, a) \notin R_3$. So, R_3 is not transitive on A .

Illustration 5

Prove that the relation R on the set $\mathbb{Z}$ of all integers defined by $(x, y) \in R \Leftrightarrow x - y$ is divisible by n is an equivalence relation on $\mathbb{Z}$.

Solution:

We observe the following properties

Reflexivity: For any $a \in \mathbb{N}$, we have

$$\Rightarrow a - a = 0 = 0 \times n \Rightarrow a - a \text{ is divisible by } n \Rightarrow (a, a) \in R$$

Thus, $(a, a) \in R$ for all $a \in \mathbb{Z}$

So, R is reflexive on $\mathbb{Z}$

symmetry: Let $(a, b) \in R$. Then,

$$\Rightarrow (a, b) \in R \Rightarrow (a - b) \text{ is divisible by } n$$

$$\Rightarrow a - b = np \text{ for some } p \in \mathbb{Z}$$

$$\Rightarrow b - a = n(-p)$$

$$\Rightarrow b - a \text{ is divisible by } n \quad [\because p \in \mathbb{Z} \Rightarrow -p \in \mathbb{Z}]$$

$$\Rightarrow (b, a) \in R$$

Thus, $(a, b) \in R \Rightarrow (b, a) \in R$ for all $a, b \in \mathbb{Z}$

So, R is symmetric on $\mathbb{Z}$.

Transitivity: Let $a, b, c \in \mathbb{Z}$ such that $(a, b) \in R$ and $(b, c) \in R$. Then,

$$(a, b) \in R \Rightarrow (a - b) \text{ is divisible by } n$$

$$\Rightarrow a - b = np \text{ for some } p \in \mathbb{Z}$$

$$(b, c) \in R \Rightarrow (b - c) \text{ is divisible by } n$$

$$\Rightarrow b - c = nq \text{ for some } q \in \mathbb{Z}$$

$$\therefore (a, b) \in R \text{ and } (b, c) \in R$$

$$\Rightarrow a - b = np \text{ and } b - c = nq$$

$$\Rightarrow (a - b) + (b - c) = np + nq$$

$$\Rightarrow a - c = n(p + q)$$

$$\Rightarrow a - c \text{ is divisible by } n \quad [\because p, q \in \mathbb{Z} \Rightarrow p + q \in \mathbb{Z}]$$

$$\Rightarrow (a, c) \in R$$

thus, $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$ for all $a, b, c \in \mathbb{Z}$. so, R is transitive relation in $\mathbb{Z}$.

Illustration 6

Show that the relation is 'congruent to' on the set of all triangles in a plane is an equivalence relation.

Solution:

Let S be the set of all triangles in a plane and let R be the relation on S defined by $(\Delta_1, \Delta_2) \in R \Leftrightarrow$ triangle Δ_1 is congruent to triangle Δ_2 . We observe the following properties.

(i) Reflexivity:

For each triangle $\Delta \in S$, we have

$$\Delta \cong \Delta \Rightarrow (\Delta, \Delta) \in R \text{ for all } \Delta \in S \Rightarrow R \text{ is reflexive on } S$$

(ii) Symmetry:

$$\text{Let } \Delta_1, \Delta_2 \in S \text{ such that } (\Delta_1, \Delta_2) \in R. \text{ Then, } (\Delta_1, \Delta_2) \in R \Rightarrow \Delta_1 \cong \Delta_2 \Rightarrow \Delta_2 \cong \Delta_1 \Rightarrow (\Delta_2, \Delta_1) \in R$$

So, R is symmetric on S

(iii) Transitivity:

$$\text{Let } \Delta_1, \Delta_2, \Delta_3 \in S \text{ such that } (\Delta_1, \Delta_2) \in R \text{ and } (\Delta_2, \Delta_3) \in R. \text{ Then,}$$

$$(\Delta_1, \Delta_2) \in R \text{ and } (\Delta_2, \Delta_3) \in R \Rightarrow \Delta_1 \cong \Delta_2 \text{ and } \Delta_2 \cong \Delta_3 \Rightarrow \Delta_1 \cong \Delta_3 \Rightarrow (\Delta_1, \Delta_3) \in R$$

So, R is transitive on S .

Hence, R being reflexive, symmetric and transitive, is an equivalence relation on S .

Illustration 7

Prove that the relation R in the set $\mathbb{N}$ of natural numbers defined by $xRy \Leftrightarrow (x - y)$ is divisible by 5 is an equivalence relation.

Solution:

For proving the relation R an equivalence relation we shall prove that the relation R is reflexive, symmetric and transitive in the set $\mathbb{N}$.

(i) Reflexive :

Let $x \in \mathbb{N}$, then

$$\Rightarrow x \in \mathbb{N} \Rightarrow x - x = 0 \Rightarrow x - x = 0 \times 5$$

$$\Rightarrow x - x \text{ is divisible by } 5 \Rightarrow x R x$$

$\therefore R$ is reflexive.

(ii) Symmetric :

Let xRy & $x, y \in \mathbb{N}$, then

$$xRy \Rightarrow x - y \text{ is divisible by } 5$$

$$\Rightarrow x - y = 5 \times p \text{ where } p \in I$$

$$\Rightarrow y - x = 5 \times (-p) \Rightarrow y - x \text{ is divisible by } 5 \quad [\because p \in I \Rightarrow -p \in I]$$

$$\Rightarrow y R x$$

$\therefore R$ is symmetric.

(iii) Transitive :

Let xRy and yRz then

$$xRy \Rightarrow x - y \text{ is divisible by } 5 \Rightarrow x - y = 5 \times p, p \in I$$

$$\text{and } yRz \Rightarrow y - z = 5q; (\because q \in I)$$

$$\Rightarrow (x - y) + (y - z) = 5p + 5q \Rightarrow x - z = 5(p + q)$$

$$\Rightarrow x - z \text{ is divisible by } 5 [\because p \in I, q \in I \Rightarrow p + q \in I]$$

$$\Rightarrow xRz$$

$\therefore R$ is transitive.

Thus R is reflexive, symmetric and transitive therefore R is an equivalence relation on the set N .

Illustration 8

In the set $\mathbb{N} \times \mathbb{N}$ consider the relation R defined as $(a, b)R(c, d) \Leftrightarrow ad = bc; a, b, c, d \in \mathbb{N}$

Show that this is an equivalence relation.

Solution:

For proving the relation R on $N \times N$ to be an equivalence relation, we will prove that the relation R is reflexive, symmetric and transitive.

Reflexive : Let $x, y \in N$, then

$$\Rightarrow (x, y) \in N \times N \Rightarrow xy = yx, \text{ by commutativity in } N$$

$$\Rightarrow (x, y)R(x, y) \forall (x, y) \in N \times N$$

$\therefore R$ is reflexive

Symmetric : Let $(x, y)R(p, q)$ where $x, y, p, q \in N$, then

$$(x, y)R(p, q) \Rightarrow xq = yp \Rightarrow yp = xq$$

$$\Rightarrow py = qx \quad [\text{by commutativity in } N]$$

$$\Rightarrow (p, q)R(x, y)$$

$\therefore R$ is symmetric.

Transitive :

Relations

Let $(x_1, y_1)R(x_2, y_2); (x_2, y_2)R(x_3, y_3)$ where

$\Rightarrow (x_1, y_1), (x_2, y_2), (x_3, y_3) \in N \times N$, then

$$\Rightarrow (x_1, y_1)R(x_2, y_2) \Rightarrow x_1y_2 = y_1x_2 \quad \dots(1)$$

$$\therefore (x_2, y_2)R(x_3, y_3) \Rightarrow x_2y_3 = y_2x_3 \quad \dots(2)$$

Now $x_1y_2 = y_1x_2$

$$x_1y_2 = y_1 \frac{y_2x_3}{y_3} \quad (\text{putting the value of } x_2 \text{ from (2)})$$

$$\Rightarrow x_1y_3 = y_1x_3 \quad \dots(3)$$

$$\Rightarrow (x_1, y_1)R(x_3, y_3)$$

Therefore, R is transitive.

Hence the relation defined on $N \times N$ is an equivalence relation.

Illustration 9

Let the relation R be defined in the set of real number as follows :

$$(i) aRb \Leftrightarrow a \geq b \quad (ii) aRb \Leftrightarrow |a| = |b|$$

Which of the above defined relation are equivalence relation.

Solution:

Let A be the set of real numbers

Reflexive : If $a \in A$, then

$$\Rightarrow a \in A \Rightarrow a \geq a \Rightarrow aRa$$

$\therefore R$ is reflexive.

Symmetric : If aRb , then

$$\Rightarrow aRb \Rightarrow a \geq b \Rightarrow b = a \text{ but } b \neq a$$

does not imply $b \geq a$ does not imply bRa

$\therefore R$ is not symmetric.

Transitive : If aRb & bRc , then

$$aRb \text{ and } bRc \Rightarrow a \geq b \text{ and } b \geq c$$

$$\Rightarrow a \geq c \Rightarrow aRc$$

$\therefore R$ is transitive.

Hence R is reflexive and transitive but not symmetric.

Thus, R is not an equivalence relation.

(ii) Reflexive: If $a \in A$, then

$$\Rightarrow aRa \Rightarrow |a| = |a|$$

R is symmetric.

Symmetric : If aRb , then

$$aRb \Rightarrow |a| = |b| \Rightarrow |b| = |a| \Rightarrow bRa$$

$\therefore R$ is symmetric.

Transitive : If aRb and bRc , then

$$aRb \text{ and } bRc \Rightarrow |a| = |b| \text{ and } |b| = |c|$$

$$\Rightarrow |a| = |c| \Rightarrow aRc$$

$\therefore R$ is transitive.

$\therefore R$ is reflexive, symmetric and transitive.

Hence R is an equivalence relation.